

30.08.26

Test

1. Even no. from 1 to 50

```
for (let i = 1; i <= 50; i++) {
```

```
  if (i % 2 === 0)
```

```
    console.log(i);
```

```
}
```

2. Factorial using while Loop.

```
const number = 5;
```

```
let fact = 1;
```

```
let i = 1;
```

```
while (i <= number) {
```

```
  fact *= i;
```

```
  i++;
```

```
}
```

```
console.log('Fact of ${number} is ${fact}');
```

output: Fact 5 is 120

3. Numbers from 20 to 1 using do..while.

```
let i = 20;
```

```
do {
```

```
  console.log(i);
```

```
  i--;
```

```
} while (i >= 1);
```

4. Reverse "JavaScript" using a loop.

```
const str = "JavaScript";
let reversed = "";
for (let i = str.length - 1; i >= 0; i--) {
    reversed += str[i];
}
console.log(reversed);
```

Output:

tpircSavaJ

5. Find the highest No:

```
const numbers = [45, 12, 89, 34, 67, 90, 23];
```

```
let highest = numbers[0];
```

```
for (let i = 1; i < numbers.length; i++) {
```

```
    if (numbers[i] > highest) {
```

```
        highest = numbers[i];
```

```
    }
```

```
}
console.log("Highest No:", highest);
```

output:

Highest Number: 90

6. Print odd numbers & find their total.

```
const numbers = [10, 15, 20, 25, 30, 35, 40];
```

```
let total = 0;
```

```
for (const number of numbers) {
```

```
    if (number % 2 !== 0) {
```

```
        console.log(number);
```

```
        total += number;
```

```
    }
```

```
}
console.log("Total:", total);
```

output:

15

25

35

Total: 75

7. Student object:

```
const student = {  
  name: "Ravi",  
  age: 22,  
  course: "JavaScript",  
  mark: 85  
};  
  
console.log("Name:", student.name);  
console.log("Age:", student.age);  
console.log("Course:", student.course);  
console.log("Mark:", student.mark);
```

8. Students who scored more than 75.

```
const students = [  
  { name: "Arun", mark: 85 },  
  { name: "Bala", mark: 65 },  
  { name: "Charan", mark: 90 },  
  { name: "Deepak", mark: 70 }  
];
```

```
for (const student of students) {
```

```
  if (student.mark > 75) {  
    console.log(student.name, student.mark);  
  }  
}
```

output:

Arun 85
Charan 90

9. Three ways to calculate square:

i) Normal Function:

```
function square (number) {  
  return number * number;  
}  
console.log (square(5));
```

ii) Function Expression:

```
const squareExpression = function (number) {  
  return number * number;  
};  
console.log (squareExpression(5));
```

iii) Arrow Function:

```
const squareArrow = (number) => number * number;  
console.log (squareArrow(5));
```

output:

25

10. Template Literal Function

```
function getMessage (name, age) {  
  return `My name is ${name} and I am ${age} years old.`;  
}  
console.log (getMessage("Ravi", 25));
```

output:

My name is Ravi & I am 25 years old.

11. var, let and const block scope

```
var a = 10;
```

```
{
```

```
  var a = 20;
```

```
  let b = 30;
```

```
  const c = 40;
```

```
  console.log("Inside block:");
```

```
  console.log("var:", a);
```

```
  console.log("let:", b);
```

```
  console.log("const:", c);
```

```
}
```

```
console.log("outside block:");
```

```
console.log("var:", a);
```

output: Inside block:

```
var: 20
```

```
let: 30
```

```
const: 40
```

outside block:

```
var: 20
```

12. Predict the output:

```
console.log(a);
```

```
var a = 10;
```

output:

undefined

13. const student = {

```
  name: "Ravi",
```

```
  age: 22,
```

```
  course: "Java Script"
```

```
};
```

```
const { name, course } = student;
```

```
console.log("Name:", name);
```

```
console.log("Course:", course);
```

output: Name: Ravi

course: JavaScript

14. Arrow function with Rest parameter.

```
const calculateTotal = (...numbers) => {  
  return numbers.reduce((total, number) => total +  
    number, 0);  
};
```

```
console.log(calculateTotal(10, 20, 30, 40));  
output : 100
```

15. Display name, city, Marks

```
const users = [  
  {  
    name: "Ravi",  
    marks: [80, 90, 85],  
    address: {  
      city: "Chennai"  
    }  
  },  
  {  
    name: "Kumar",  
    marks: [60, 70, 75]  
  }  
];
```

output:

```
Name: Ravi  
City: Chennai  
Total Marks: 255  
- - - - -  
Name: Kumar  
City: City Not Available  
Total Marks: 205.
```

```
for (const { name, marks, address } of users)  
  const city = address?.city ?? "City Not Available";  
  const totalMarks = marks.reduce((total, mark) =>  
    total + mark, 0);  
  
  console.log("Name:", name);  
  console.log("City:", city);  
  console.log("Total Marks:", totalMarks);  
  console.log(" - - - - -");
```